CS 372 Final Project — Self-Assessment

Introduction to Applied Machine Learning

Student & Project Information

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Project title	From Landscape to Biomass
Repository link	https://github.com/jamess1000/CS372_Final_Project

How to Use This Template

Fill out this template in Word (or Google Docs → download as .docx), then upload the completed .docx to Gradescope alongside your repository link. Please do not convert to PDF — the editable .docx format is what we are expecting.

Filling in evidence pointers

For every item you claim, provide an evidence pointer precise enough that a grader can locate your work quickly. The handout asks for short pointers like:

- data augmentation in src/train.py lines 45–67
- ablation study results in notebooks/experiments.ipynb
- guardrails discussion in README.md Evaluation section
- architecture explanation in technical walkthrough 4:30–6:15

You do not need to describe what you did in detail — just point us where to look.

Important Rules

- Category 1 (Machine Learning) is capped at 15 claims. The Category 1 table below has exactly 15 slots; fill in as many as apply, leaving the rest blank if you claim fewer than 15.
- Stackable add-ons (marked "+3 pts, stackable" in the handout) each count as one of your 15 Category 1 claims.
- For tiered items (Used vs. Modified/Adapted vs. Developed), claim only the tier that matches your work — do not claim both tiers of the same item.
- One piece of work, one item. If your work fits multiple rubric items, claim the most specific or highest-value one.
- "Evidence of impact" items require a quantitative or qualitative comparison (e.g., metrics with/without the technique, a before/after chart, or a side-by-side with discussion). Your evidence pointer should lead to that comparison.
- Categories 2 and 3 have no cap — claim every item that applies. Check the box by replacing ☐ with ☒ (or just type an X next to it).

Category 1: Machine Learning (max 73 pts)

Claim up to 15 items from the Machine Learning rubric in the handout. For each claim, write the rubric item as it appears in the handout (short paraphrase is fine), the point value, and a specific evidence pointer. Leave unused slots blank.

#	Rubric item claimed (copy from handout)	Points	Evidence pointer
1	Successfully adapted a pretrained model across substantially different domains or tasks	7	Transfer learning with a pretrained ResNet in models/CustomCNN_and_Resnet_models_fin.ipynb
2	Modified/Adapted a vision transformer by fine-tuning on dataset	7	Fine-tuned ViT-B/16 in mpdels/vit_models_fin.ipynb
3	Used a vision transformer for inference or as a frozen backbone	5	Custom Vision Transformer from Scratch in models/vit_models_fin.ipynb
4	Built multi-stage ML pipeline connecting outputs of one model to inputs of another	7	The modeling done in notebooks/Predict_Feature_Plus_Tabular.ipynb
5	Compared multiple model architectures or approaches quantitatively with controlled experimental setup	7	Easiest to see in Overall comparison bar chart in notebooks/model_comparison_fin.ipynb, but present in discussions throughout all notebooks
6	Conducted ablation study systematically varying at least two independent design choices…	7	Modality ablation summary in notebooks/model_comparison_fin.ipynb
7	Defined and trained a custom neural net architecture using Pytorch	5	Many times, easiest to see in Custom CNN from Scratch in models/ CustomCNN_and_Resnet_models_fin.ipynb
8	Trained model using GPU/TPU/CUDA acceleration	3	Essentially every model, seen in moving to device at the top of each notebook (ie: models/vit_models_fin.ipynb)
9	Applied regularization techniques to prevent overfitting	5	Train the custom CNN in models/ CustomCNN_and_Resnet_models_fin.ipynb. Note that we use dropout and early stopping with patience
10	Created a baseline model for comparison	3	Sanity-floor-baseline in models/vit_models_fin.ipynb

#	Rubric item claimed (copy from handout)	Points	Evidence pointer
11	Applied feature engineering	5	Feature engineering in models/tabular_models_fin.ipynb
12	Tracked and visualized training curves showing loss and/or metrics over time	3	Train the Fusion model in models/combined_model_fin.ipynb
13	Designed and implemented a custom architecture combining elements from multiple paradigms	10	The entire notebook in models/combined_model_fin.ipynb, in particular note the Fusion model architecture section.
14	Modular code design with reusable	3	Code design of all notebooks, in particular note the code design of models/ CustomCNN_and_Resnet_models_fin.ipynb
15	Used appropriate data loading with batching and shuffling	3	Pytorch Dataset and Dataloader / Build DataLoaders for each track in models/ CustomCNN_and_Resnet_models_fin.ipynb

Category 2: Following Directions (max 15 pts)

Claim every item that applies.

Submission and Self-Assessment

Claim	Pts	Rubric item
X	3	Self-assessment submitted following guidelines (≤15 ML items with evidence)

Documentation (1 pt each)

Claim	Pts	Rubric item
X	1	SETUP.md with clear, step-by-step installation instructions
X	1	ATTRIBUTION.md with detailed attribution including AI generation info
X	1	requirements.txt or environment.yml present and accurate

Claim	Pts	Rubric item
X	1	README.md: What it Does section (one paragraph)
X	1	README.md: Quick Start section
X	1	README.md: Video Links section with direct links
X	1	README.md: Evaluation section
X	1	README.md: Individual Contributions section (group projects)

Video Submissions (2 pts each)

Claim	Pts	Rubric item
X	2	Demo video — correct length, appropriate for non-specialist audience, no code shown
X	2	Technical walkthrough — correct length, clearly explains code, ML techniques, and key contributions

Project Workshop Days (1 pt each)

Claim	Pts	Rubric item
☐	1	Attended first project workshop day
X	1	Attended second project workshop day
☐	1	Attended third project workshop day
☐	1	Attended fourth project workshop day

Category 3: Project Cohesion and Motivation (max 15 pts)

Claim every item that applies.

Project Purpose and Motivation (3 pts each)

Claim	Pts	Rubric item
X	3	README clearly articulates a single, unified project goal or research question
X	3	Demo video effectively communicates why the project matters to a non-technical audience
X	3	Project addresses real-world problem or meaningful research question (connected to concrete papers or competitions)

Technical Coherence (2 pts each)

Claim	Pts	Rubric item
X	2	Technical walkthrough demonstrates components working together synergistically (not just isolated experiments)
X	2	Project shows clear progression from problem → approach → solution → evaluation
X	2	Design choices explicitly justified in videos or documentation
X	2	Evaluation metrics directly measure the stated project objectives
X	2	No superfluous 'point collecting' — all major ML components serve the larger project goals
X	2	Clean codebase with readable code and no extraneous, stale, or unused files

Totals

Category	Points Claimed
Machine Learning	73 / 73
Following Directions	15 / 15

Category	Points Claimed
Project Cohesion and Motivation	15 / 15
Grand Total (can be up to 103)	103 / 100

Self-Reflection (Optional but Encouraged)

These questions are not graded. They are a chance to reflect on your work, and your answers help us understand your project. A few sentences per question is plenty.

1. What are you most proud of in this project?

The collaboration and teamwork throughout this project, and the ability to apply several different architectures we learned throughout the course of this class to a single problem. Not all of them worked as well as we had hoped, but it was useful experience in learning how to design custom architectures for a real-world problem.

2. What was the hardest technical challenge, and how did you work through it?

The hardest technical challenge was limiting architectures to vision-only. In essence, the competition we chose requires that the submitted model only use vision and no tabular features, but tabular models tended to work better. To work through this, we explored multiple different vision architectures, combination architectures, etc. Ultimately, we think that the tabular data is useful in the real-world (outside of this individual competition) in the sense that many of the features are information farmers would know at the time of estimation, thus it is not unreasonable to create these models that incorporate them since it would still have real application.

3. If you had another two weeks on this project, what would you do next?

If I had another two weeks, we'd explore more powerful pre-trained vision transformers for transfer learning and work to find hidden representations of the tabular features that the GBM deems as highly important that are extractable from the image alone.